

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – III
PAPER –2
TEST DATE: 31-07-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

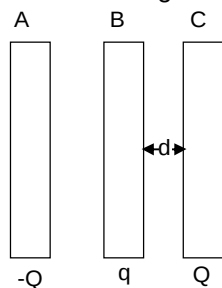
Physics

PART – I

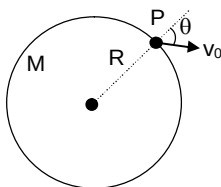
Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. Two fixed very large conducting plates A and B of area S carry charges $-Q$ and q respectively where $Q > q > 0$. A third identical plate C carrying a charge Q is released at distance d from B. Third plate collides with B. Assume collision is elastic and time of collision is sufficient to redistribute charge among B and C (Neglect gravity):



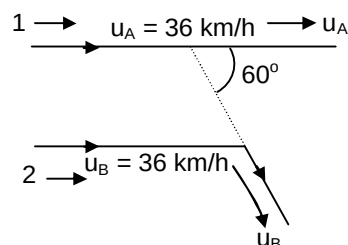
- (A) Force acting on plate C before collision is $\frac{(Q-q)Q}{2\epsilon_0 S}$.
- (B) Force acting on plate C after collision is $\frac{(q/2)^2}{2\epsilon_0 S}$.
- (C) Subsequent to collision kinetic energy of plate C increases.
- (D) Subsequent to collision electrostatic potential energy between the plates remains constant.
2. Consider a planet of mass M and radius R as shown in the figure. a particle is projected from the surface of the planet at an angle θ from radial direction and with velocity equal to the orbital velocity of circular orbit at that point then choose the correct statement(s):



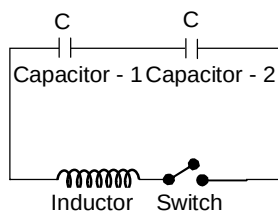
- (A) Maximum distance of particle from centre of the planet is $(R + R\cos\theta)$.
- (B) Path of the particle is parabola.
- (C) Time taken by the particle to return to the surface of planet is $\sqrt{\frac{R^3}{GM}}$.
- (D) Time taken by the particle to reach the surface of planet is equal to $\pi\sqrt{\frac{R^3}{GM}}$.
3. Highly excited states for hydrogen like atoms (also called Rydberg states) with nuclear charge Ze are defined by their principal quantum number n , where $n \gg 1$. Which of the following statement(s) is (are) true?
- (A) Relative change in the radii of two consecutive orbitals does not depend on Z
- (B) Relative change in the radii of two consecutive orbitals varies as $\frac{1}{n}$

- (C) Relative change in the energy of two consecutive orbitals varies as $\frac{1}{n^3}$
- (D) Relative change in the angular momentum of two consecutive orbitals varies as $\frac{1}{n}$

4. Two trains move in the same direction on two close tracks. The train A sounds a horn of single frequency of 1 kHz. At a certain instant of time, when both trains move at the same speed of 36 km/hr, the two straight tracks deviate at an angle of 60° with each other. Velocity of sound in air is 300 m/s. Which of the following statement (s) is/are true?



- (A) The apparent frequency heard by a passenger on the second train B just after he passes the bend will be 1.05 kHz.
- (B) The apparent frequency heard by a passenger on the first train A, just after the second train B passes the bend will be 1 kHz.
- (C) The apparent frequency heard by a passenger on the second train B just after he passes the bend will be 983 Hz.
- (D) If the first train A, that produces the sound, were to pass the bend while the other train B goes on the straight track then the apparent frequency heard by a passenger on the latter train B will be less than 1 kHz.
5. Consider the circuit shown with respective specifications of elements marked in the figure. Capacitor - 1 is charged such that charge on it is Q_0 while capacitor - 2 is uncharged. The switch is closed at $t = 0$.



- (A) Frequency of oscillation of charge on left plate of capacitor - 1 is $\frac{1}{\pi\sqrt{2LC}}$
- (B) Frequency of oscillation of charge on left plate of capacitor-1 is $\frac{1}{\pi}\sqrt{\frac{2}{LC}}$
- (C) Maximum current through the inductor is $\frac{Q_0}{\sqrt{2LC}}$
- (D) Maximum current through the inductor is $\frac{Q_0}{\sqrt{LC}}$
6. A lens forms 3 times magnified image of an object on screen. When the distance between the screen and the lens is increased by 10 cm and object is also moved to get 5 times magnified image again on the screen. Then
- (A) Focal length of the lens is 5 cm
- (B) Focal length of the lens is - 5 cm
- (C) The displacement of the object is $\frac{2}{3}$ cm towards the lens
- (D) The displacement of the object is $\frac{2}{3}$ cm away the lens

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

You may have heard of g – LOC. In simple terms, it is gravity induced loss of consciousness. This seems surprising. How can gravity lead to a loss of consciousness. For this, we have to remember that brain functions properly when the heart maintains a proper blood pressure in the brain. The brain is higher than heart by about 30 cm, so blood pressure at brain is less than that at the heart. Where there is a sudden change in acceleration, the pressure at heart changes. If this decreases the pressure in brain, the brain system can stop functioning. This decrease in pressure can be fatal. It is seen that when the effective acceleration due to gravity is 3g, the person can undergo g – LOC. For this problem, we can assume that density of blood = $1 \times 10^3 \text{ kg/m}^3$.

7. Assume that a pilot flying a plane takes a horizontal turn with a speed of 100 m/s. What is the minimum radius he can take?
 - (A) $\frac{1000}{3} \text{ m}$
 - (B) $250\sqrt{2} \text{ m}$
 - (C) 250 m
 - (D) $\frac{250}{3} \text{ m}$
8. What is the pressure difference between heart and brain when there is g – LOC?
 - (A) 1000 Pa
 - (B) 3000 Pa
 - (C) 6000 Pa
 - (D) 9000 Pa

Paragraph for Question Nos. 09 and 10

This problem investigates the Planck black body spectrum in one dimension. Consider a one – dimensional electromagnetic cavity of length L. At the ends of cavity, there are perfect reflectors. The waves form standing wave pattern in cavity analogous to a string fixed at both ends. Thus in frequency range ν to $\nu + d\nu$, we have number of modes as dN which can be found by formulae for standing waves in a string. Power stored in interval ν to $\nu + d\nu$ is given by $\bar{E}(\nu)dN$. We can assume

$$\int_0^\infty \frac{x}{e^x - 1} dx = \frac{\pi^2}{6}. \text{ Theoretically all frequencies from } 0 \text{ to } \infty \text{ are possible.}$$

The average energy in a single mode of frequency ν is given by $\bar{E}(\nu) = \frac{h\nu}{e^{h\nu/kT} - 1}$. Use this to write down the one – dimensional black body spectrum. This is the energy stored in the frequency interval $(\nu + d\nu)$. In three dimensions, Stefan found that the total power radiated by a black body was proportional to T^4 .

9. The number of modes in frequency range ν to $\nu + d\nu$:
 - (A) Increase with increasing length L
 - (B) Decrease with increasing length L
 - (C) Increase with increases in number of harmonic
 - (D) Decrease with increase in number of harmonic

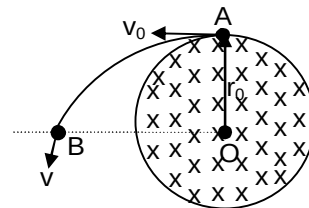
10. Power P is proportional to:

- (A) T^4
 (B) T^2
 (C) L^{-3}
 (D) L^2

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. A graph of $\sqrt{\nu}$ (where ν is the frequency of K_α line of the characteristic X – ray spectrum) is plotted against the atomic number Z of the elements emitting the characteristic X – ray. The intercept of the graph on the z – axis is 1 and the slope of the graph is 0.5×10^8 S.I. units. The frequency of the K_α line for an element of atomic number 41 is given as $\alpha \times 10^{18}$ Hz. Find the value of α .
12. The outer surface of a transparent glass slab of refractive index $\mu_1 = 1.5$ is coated by a thin layer of transparent medium of refractive index $\mu_2 = 1.6$. Orange light of wave length $6400 \text{ \AA}$ fall normally on the coat. The reflected light at the upper surface and at the lower surface of the coat interfere destructively. If thickness of the coat is $5K \times 10^{-8} \text{ m}$, calculate the minimum value of K .
13. Consider a cylindrical region of uniform but time varying magnetic field. Now consider a plane perpendicular to this field. Top view of this cylindrical magnetic field on the plane is shown in the diagram which is a circular region of radius $r_0 = 1 \text{ m}$. A particle of mass $\pi \text{ kg}$ and charge 2 C is projected with velocity of $v_0 = \sqrt{5} \text{ m/s}$, tangential to this circle from point A. This particle intersects another radial line OB at point B with velocity v . If the magnetic field is increasing at the rate of 4 T/s then find the value of v .



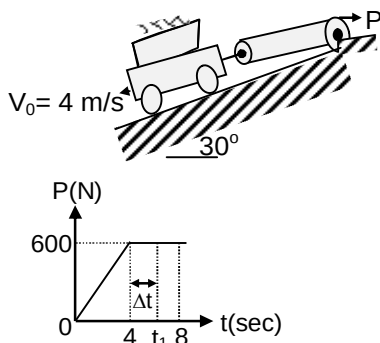
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

The loaded 150 kg skip is rolling down the incline at 4 m/s when a force P is applied to the cable as shown at time $t = 0$. The force P is increased uniformly with the time until it reaches 600 N at $t = 4 \text{ s}$, after which it remains constant at this value.



14. The time t_1 at which the skip reverses its direction is nearly _____ sec.
15. The velocity v (in m/s) of the skip at $t = 8$ s (treating the skip as a particle) is :

Question Stem for Question Nos. 16 and 17

Question Stem

A rod AB of mass M and length L is lying on a horizontal frictionless surface. A particle of mass m travelling along the surface hits the end A of the rod with a velocity v_0 in a direction perpendicular to AB. The collision is completely elastic. After the collision the particle comes to rest:

16. Find the ratio $\frac{M}{m}$
17. Angle turned by rod in time $\left(\frac{\pi L}{3v_0} \right)$,

Question Stem for Question Nos. 18 and 19

Question Stem

A 500 g teapot and an insulated thermos are in a 20°C room. The teapot is filled with 1000 g of the boiling water. 12 tea bags are then placed into the teapot. The brewed tea is allowed to cool to 80°C , then 250 g of the tea poured from the teapot into the thermos. The teapot is then kept on an insulated warmer that transfers 500 cal/min to the tea. Assume that the specific heat of brewed tea is the same as that of pure water, and that the tea bags have a very small mass compared to that of the water. And a negligible effect on the temperature, the specific heat of teapot is 0.17 J/g K and that of water is 4.18 J/g K . The entire procedure is done under atmospheric pressure there are 4.18 J in one calorie.

18. After the tea is added to the thermos, the temperature of the liquid quickly falls from 80°C to 75°C as it reaches thermal equilibrium with the thermos flask. What is the heat capacity (in J/K) of the thermos?
19. If after some of the tea has been transferred to the thermos (as described in the passage), the teapot with its contents (at a temperature of 80°C) was placed on the insulated warmer for 5 minutes, what would be the temperature (in $^\circ\text{C}$) at the end of this 5 minute period? (Assume that no significant heat transfer occurs with the surroundings)

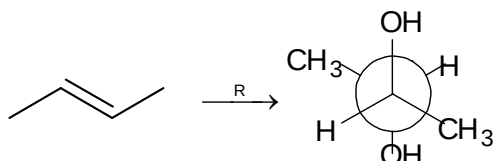
Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

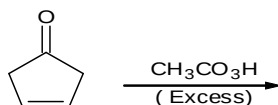
20.



R may not be

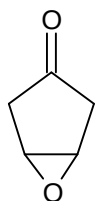
- (A) $\text{HCOOH}/\text{H}_2\text{O}_2$ (35%)
- (B) $\text{OsO}_4/\text{H}_2\text{O}_2$
- (C) Cold MnO_4^-
- (D) CH_3COOH

21.

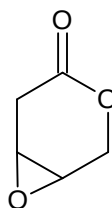


Product of reaction can not be:

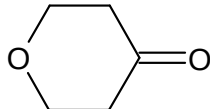
(A)



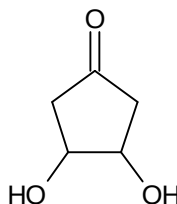
(B)



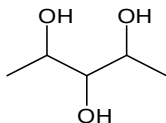
(C)



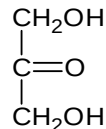
(D)

22. Which will undergo oxidation by IO_4^- ?

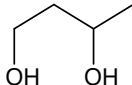
(A)



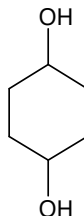
(B)



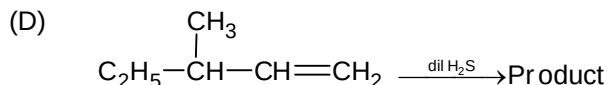
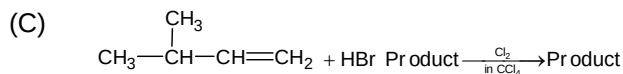
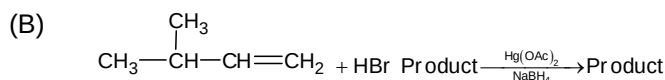
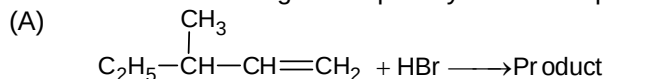
(C)



(D)



23. In which of the following case optically active compound is produced (major)?



24. 1 mole of a non-volatile non-dissociative solid solute is dissolved in 200 moles of water. The solution is taken to a temperature T_K (lower than freezing point of solution) to cause ice formation. After removal of ice the remaining solution is taken to 373 K where vapour pressure is observe to be 740 mm of Hg. Identify the **correct** options:

Given Data: $K_f(\text{H}_2\text{O}) = 2 \text{ K kg mol}^{-1}$ and normal boiling point of $\text{H}_2\text{O} = 373 \text{ K}$.

(A) 163 moles of ice will be formed.

(B) Temperature of which original solution was cooled should be $\frac{2000}{37 \times 18} \text{ K}$.

(C) Freezing point of original solution should be $-\frac{10}{18}^\circ\text{C}$.

(D) Relative lowering of vapour pressure of final solution will be $\frac{1}{201}$.

25. An element A (Atomic weight = 120) having bcc structure has unit cell edge length 400 pm. Identify the correct option(s) (**Given:** $N_A = 6 \times 10^{23}$)

(A) The density of solid element is 6.25 gm/cm^3 .

(B) There are 6×10^{22} unit cells in 24 gm of the solid element.

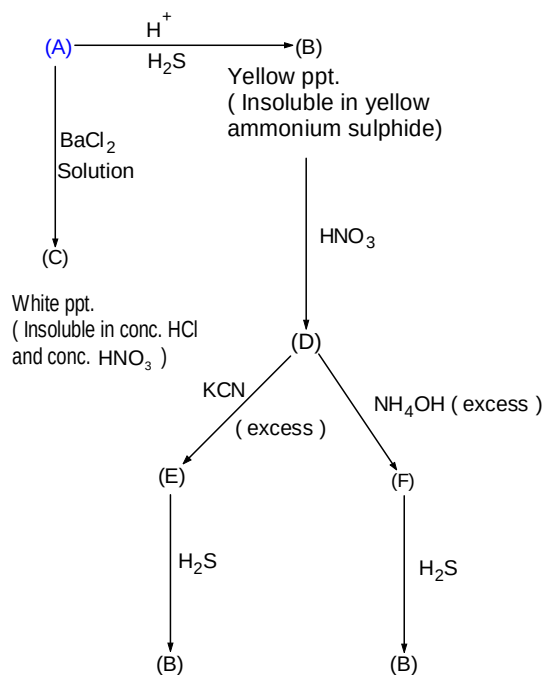
(C) The atomic radius is about $1.732 \text{ \AA}$.

(D) In 25 gm of solid element, the volume occupied by atoms is nearly 2.72 cm^3 only.

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 26 and 27



26. What is C?
 (A) PbCl_2
 (B) BaSO_4
 (C) BaSO_3
 (D) PbSO_4
27. What is B?
 (A) CdS
 (B) PbCrO_4
 (C) HgO
 (D) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$

Paragraph for Question Nos. 28 and 29

Crystal field theory was proposed by H. Bethe (1929) and Van Vleck (1932) and as originally applied to ionic crystals to explain their optical properties and is, therefore, called crystal field theory, however, this theory was applied to the study of coordination compounds in 1950 and this theory accounts for more satisfactory explanation for the properties of complexes especially colour and magnetism.

28. What is the hybridization of $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$?
 (A) sp^3d^2
 (B) d^2sp^3
 (C) sp^3d
 (D) dsp^3

29. What is the geometry of $[\text{Co}(\text{NH}_3)_6]^{2+}$?
- (A) Octahedral
 (B) Tetrahedral
 (C) Square planar
 (D) Square pyramidal

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. At 143 K, the reaction of XeF_4 with O_2F_2 produces a xenon compound Y. The total number of lone pair(s) of electrons present on the whole molecule of Y is
31. Among B_2H_6 , $\text{B}_3\text{N}_3\text{H}_6$, N_2O , N_2O_4 , $\text{H}_2\text{S}_2\text{O}_3$ and $\text{H}_2\text{S}_2\text{O}_8$ the total number of molecules containing covalent bond between two atoms of the same kind is.....
32. A weak acid (HA) is 1% ionized at 0.01 M. What is its pH when 100 mL of this acid is treated with 25 mL of 0.01 M $\text{Ca}(\text{OH})_2$?

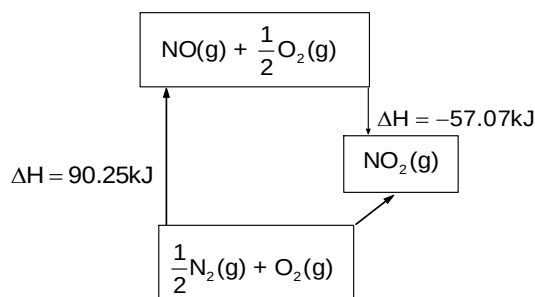
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

Question Stem

Based on the following figure:

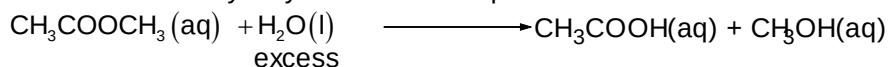


33. $\Delta_f H^\circ(\text{NO}_2)$ is _____ kJ mol^{-1} .
34. Heat of given reaction, $2\text{NO}_2 \longrightarrow 2\text{NO} + \text{O}_2$, is

Question Stem for Question Nos. 35 and 36

Question Stem

Kinetics of acidic hydrolysis of ester is a pseudo 1st order reaction



Rate law is given by : $r = K [\text{H}^+] [\text{H}_2\text{O}] [\text{CH}_3\text{COOCH}_3]$;

$K = 1.8 \times 10^{-3} \text{ M}^{-2} \text{ sec}^{-1}$ where H^+ ion concentration is given by acid catalyst. In an experiment, 1M CH_3COOH_3 is hydrolyzed using 0.1M HCl as catalyst.

35. Calculate the time (in sec) at which ester concentration reduce to 0.25M [$\ln 2 = 0.7$].
(Given : Density of pure $\text{H}_2\text{O} = 1 \text{ gm/ml}$)
36. Reaction mixture (H^+ catalyzed) is titrated against standard NaOH solution at different times. If initially 10 ml of NaOH was required while after long time 100 ml of NaOH is required then find volume (in ml) of NaOH required at 230.3 sec.

Question Stem for Question Nos. 37 and 38

Question Stem

In diamond structure, carbon atoms FCC lattice and 50% of tetrahedral voids occupied by carbon atoms. Every carbon atom is surrounded tetrahedrally by four carbon atoms with bond length 173.2 pm.

On the basis of above information answer following question ($N = 6 \times 10^{23}$)

37. The side length of diamond unit cell is: (in pm)
38. The density of diamond unit cell is: (in gm/cm^3)

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Consider two circles $C_1 : x^2 + y^2 - 1 = 0$ and $C_2 : x^2 + y^2 - 2 = 0$. Let A(1, 0) be a fixed point on the circle C_1 and B be any variable point on the circle C_2 . The line BA meets the curve C_2 again at C. Which of the following alternative (s) is /are correct ?
- (A) $OA^2 + OB^2 + BC^2 \in [7, 11]$, where O is the origin.
 (B) $OA^2 + OB^2 + BC^2 \in [4, 7]$, where O is the origin.
 (C) Locus of midpoint AB is a circle of radius $\frac{1}{\sqrt{2}}$.
 (D) Locus of midpoint AB is a circle of area $\pi/2$
40. A ray emanating from the point $(-4, 0)$ is incident on the ellipse $9x^2 + 25y^2 = 225$ at the point P with abscissa 3. Then the equation of reflected ray is/are
- (A) $5y = -12x + 48$
 (B) $5y = 12x + 48$
 (C) $5y = -5x + 3$
 (D) $5y = 5x - 3$
41. Let f be a differentiable function satisfying a relation $f(xy) = xf(y) + yf(x) - 2xy$ where $x > 0, y \in \mathbb{R}$ and $f'(1) = 3$. Identify the correct statement (s) about f(x).
- (A) $f(x) = x \ln x + 2x$
 (B) $f(x) = x^2 \ln x - 2x + 2x^2 + 2$
 (C) $x = e^{-3}$ is abscissa of the point of inflection on the graph of $y = f(x)$
 (D) curve $y = f(x)$ does not have any horizontal asymptote.
42. If in $\triangle ABC, \frac{r}{r_1} = \frac{r_2}{r_3}$ (where r is inradius and r_1, r_2, r_3 are exradii of circles opposite to vertices A, B, C respectively), then which of these statements can be true ?
- (A) $r_3 > r_2 > r_1 > r$
 (B) $r > r_1 > r_2 > r_3$
 (C) $r_3^2 = r_1 r_2 + r_2 r_3 + r_3 r_1$
 (D) $2r = R$
43. Which of the following pieces of data not uniquely determine a $\triangle ABC$ (R being the radius of circumcircle) ?
- (A) a, sin A, sin B
 (B) a, b, c
 (C) a, sin B, R
 (D) a, sin A, R

44. In R^3 , let L be a straight line passing through the origin. Suppose that all the points on L are at a constant distance from the two planes $P_1 : x + 2y - z = -1$ and $P_2 : 2x - y + z - 1 = 0$. Let M be the locus of the feet of the perpendiculars drawn from the points on L to the plane P_1 . Which of the following points lie(s) on M ?

- (A) $\left(0, \frac{-5}{6}, \frac{-2}{3}\right)$
 (B) $\left(\frac{-1}{6}, \frac{-1}{3}, \frac{1}{6}\right)$
 (C) $\left(\frac{-5}{6}, 0, \frac{1}{6}\right)$
 (D) $\left(\frac{-1}{3}, 0, \frac{2}{3}\right)$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

There is a point P inside an equilateral $\triangle ABC$ of side 'a' where distance from vertices is 3, 4, and 5 respectively. Rotate the triangle and P through 60° about C . Let A go to A' and P to P' .

45. The area of $\triangle PAP'$ is
 (A) 8
 (B) 12
 (C) 16
 (D) 6

46. $\angle APB$ is equal to
 (A) 98°
 (B) 120°
 (C) 150°
 (D) None of these

Paragraph for Question Nos. 47 and 48

Let $f : R \rightarrow R$, $y = f(x)$, $f(0) = 0$, $f'(x) > 0$ and $f''(x) > 0$. Three points $A(\alpha, f(\alpha))$, $B(\beta, f(\beta))$, $C(\gamma, f(\gamma))$ on $y = f(x)$ such that $0 < \alpha < \beta < \gamma$.

47. Which of the following is false
 (A) $\alpha f(\beta) > \beta f(\alpha)$
 (B) $\alpha f(\beta) < \beta f(\alpha)$
 (C) $\gamma f(\beta) < \beta f(\gamma)$
 (D) $\gamma (f(\alpha)) < \alpha f(\gamma)$

48. Which of following is true?
- (A) $\frac{f(\alpha) + f(\beta)}{2} < f\left(\frac{\alpha + \beta}{2}\right)$
- (B) $\frac{f(\alpha) + f(\beta)}{2} > f\left(\frac{\alpha + \beta}{2}\right)$
- (C) $\frac{f(\alpha) + f(\beta)}{2} = f\left(\frac{\alpha + \beta}{2}\right)$
- (D) $\frac{2f(\alpha) + f(\beta)}{3} < f\left(\frac{2\alpha + \beta}{3}\right)$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. If $f(x) = \sin^{-1}(x + 2019) - \cos^{-1}(x + 2021)$ and $g(x) = \sin^{-1}(x + 2017) + \cos^{-1}(x + 2019)$ and x_1, x_2 lies in domain of $f(x)$ and $g(x)$ respectively, then $\left| \lim_{x \rightarrow x_1 + x_2} \frac{\sin(x - 2x_1 - 2) - \sin(2x - 3x_2 + 2022)}{\tan(x - x_1 - x_2)} \right|$ is equal to
50. If $\sin(\sin x + \cos x) = \cos(\cos x - \sin x)$, find the largest possible value of $\frac{\pi}{\sin x}$.
51. Let $a, b, c \in \mathbb{I}^+$ such that LCM of a and b is 1000, LCM of b and c is 2000 and LCM of c and a is 2000, then number of ordered triplet (a, b, c) is X . Then largest prime which divides X is ____.

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Let A be set of all 3×3 symmetric matrices all of whose entries are either a or b or c , two of these entries are a , four of them are b and three of them are c .

52. The number of matrices A , (if a, b, c are distinct real numbers), is
53. If $a = c = 1$ & $b = 0$, then number of matrices A for which the system of linear equations

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has an unique solution, is}$$

Question Stem for Question Nos. 54 and 55**Question Stem**

Let $P_1 : 2x - 3y + 6z + 8 = 0$ and $P_2 : 3x - 2y - 2z + 7 = 0$ be two planes and $A\left(0, \frac{29}{9}, \frac{5}{18}\right)$ lies on both planes. Two points $B(2, 0, -2)$ and C are such that line of intersection of the planes is the internal angle bisector of $\angle A$ of $\triangle ABC$ and $AB = AC$.

On the basis of above information, answer the following questions.

54. If co-ordinates of C are (α, β, γ) , then $(\beta + \gamma)$ is equal to ____.
55. The x-intercept of plane containing the point C and perpendicular to planes P_1 and P_2 is ____.

Question Stem for Question Nos. 56 and 57**Question Stem**

Let 'S' be the set of first 18 natural numbers. We have to find the number of ways of selecting the numbers from 'S' in the following questions.

56. Three numbers such that they are all consecutive (or) no two of them are consecutive is
57. Two numbers such that the sum of their cubes is divisible by 3 is